

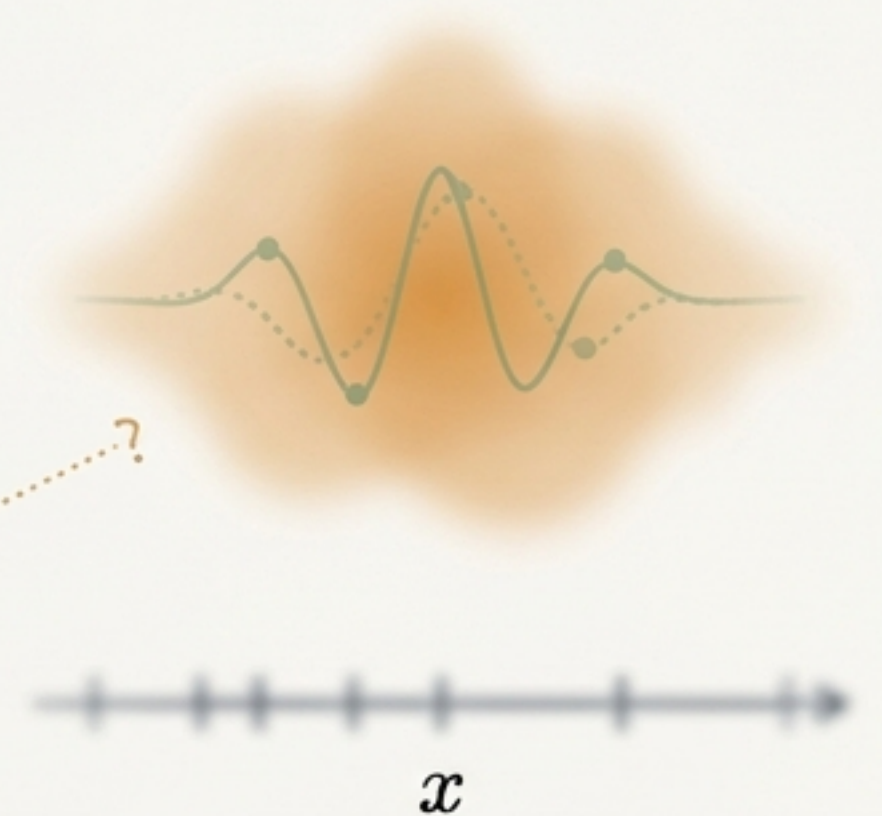
Time in Quantum Theory is an External Parameter, Not an Observable

- In both non-relativistic QM and relativistic QFT, time is treated as a **background parameter**, fundamentally different from observables like position or momentum.
- There is no universally accepted, self-adjoint time operator.
- This is not an oversight, but a consequence of deep theoretical challenges, notably Pauli's obstruction for Hamiltonians bounded from below.

Parameter: Classical Time



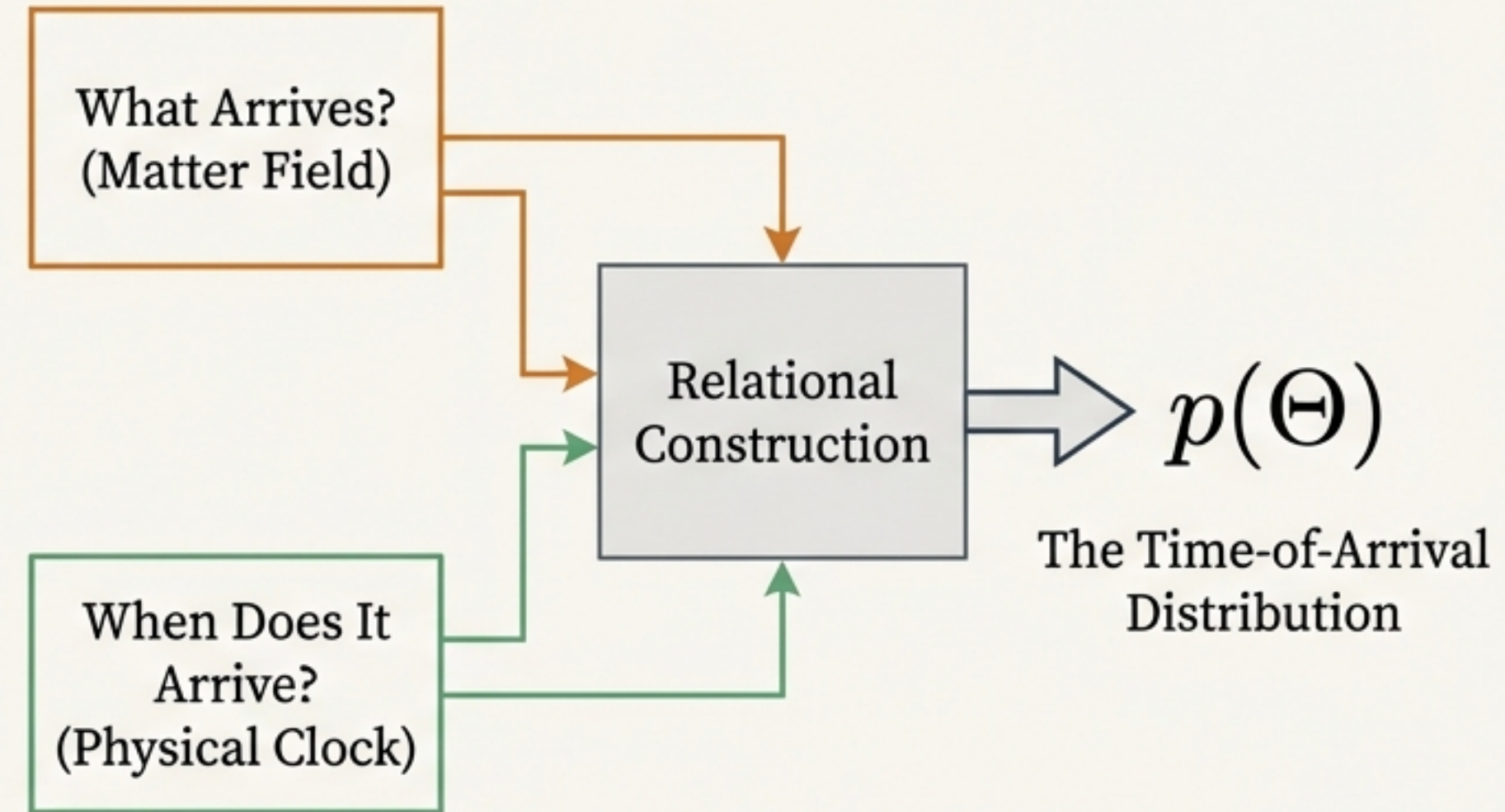
Observable: Quantum State



How, then, do we describe an operationally meaningful concept like the “time of arrival” of a particle at a detector?

Our Strategy: Build Time from Physical Fields

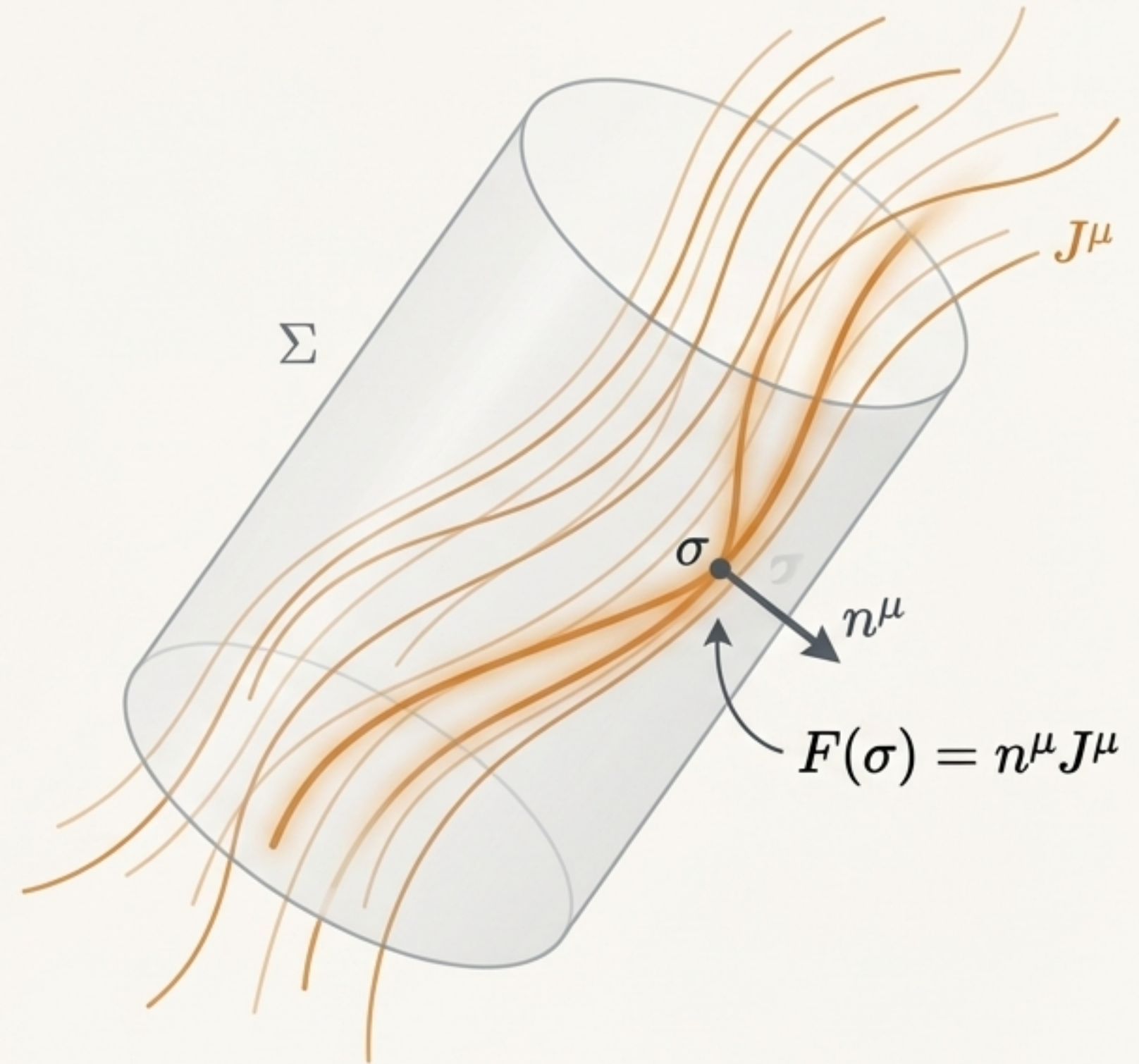
- Instead of searching for a universal time operator, we will define TOA as a **relational observable** constructed from physical fields measured at a detector.
- Time is not postulated; it **emerges from the interaction** between a particle and a measurement apparatus.
- This approach is **inherently operational** and **observer-dependent** in a controlled, physical way.
- Crucially, it **avoids Pauli-type obstructions** by never postulating a time operator conjugate to the Hamiltonian.



Pillar 1: The Matter Flux Tells Us *What* Arrives

Key Concept & Definitions

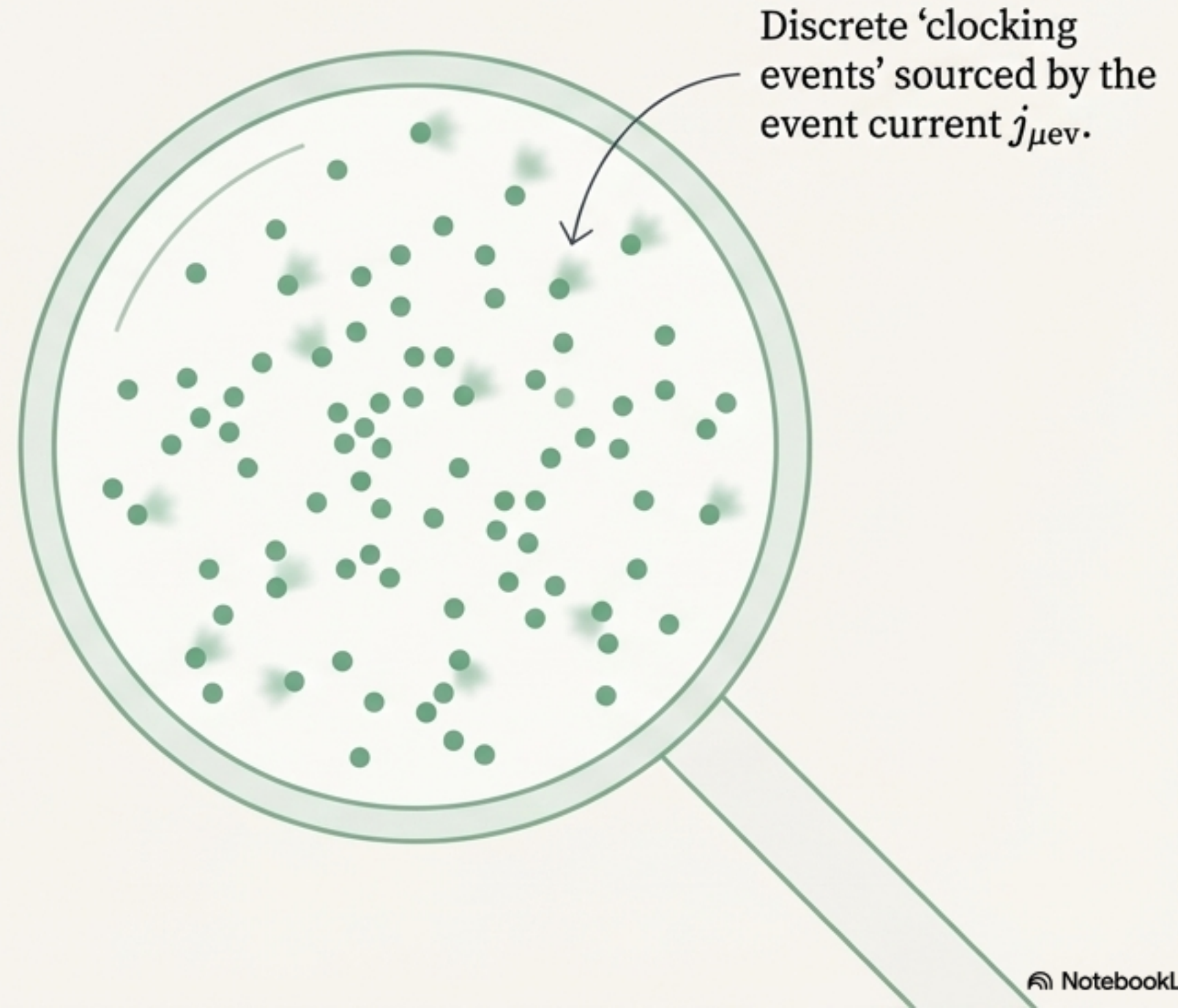
- **Key Concept:** The arrival of a particle is physically registered as a flux of a conserved **matter current**, J^μ (where $\nabla_\mu J^\mu = 0$), through the detector.
- **Detector:** Modeled as a timelike world-tube Σ in spacetime.
- **Matter Flux $F(\sigma)$:** The component of the matter current normal to the detector's surface: $F(\sigma) \equiv n^\mu J^\mu$.
- **Total Registered Flux N :** The integral of $F(\sigma)$ over the entire detector surface, $N = \int_\Sigma d^3\sigma \sqrt{|h|} F(\sigma)$.



Pillar 2: An Event Current Provides the Raw Ticks for a Physical Clock

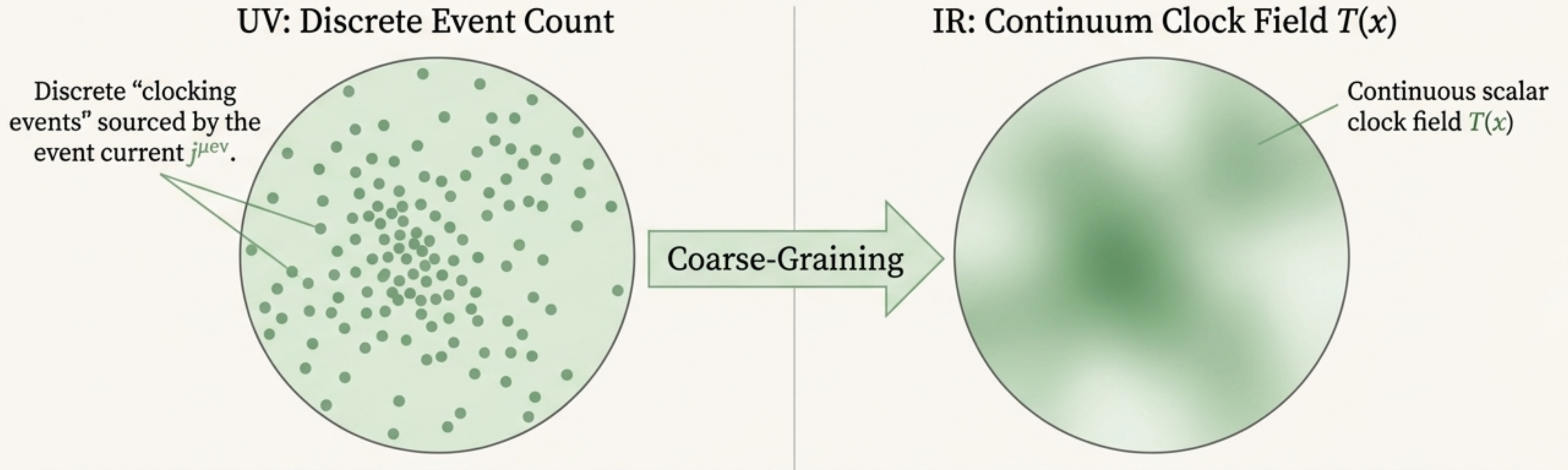
Key Concept & Definitions

- **Key Concept:** We postulate the existence of a second, distinct conserved (or approximately conserved) vector field, the event current $j^{\mu ev}$.
- **Event Current** $j^{\mu ev}$: A vector field satisfying $\nabla_\mu j^{\mu ev} \approx 0$.
- **Event Count** $N_{ev}(\Sigma)$: The total flux of this current through a surface Σ . $N_{ev}(\Sigma) \equiv \int_\Sigma d\Sigma_\mu j^{\mu ev}$.
- **The Discrete Clock** T_{disc} : The event count itself serves as a discrete, integer-valued clock. $T_{disc} \equiv N_{ev}$.
- **Interpretation:** Operationally, N_{ev} counts the number of discrete 'clocking events' that have occurred on the detector. This is the fundamental, high-resolution (UV) picture of our clock.



From Discrete Events to a Continuum Clock Field

The discrete event-count clock (T_{disc}) is the high-resolution “ultraviolet” (UV) description. By coarse-graining over a spacetime scale ℓ , we obtain a smooth “infrared” (IR) description.



The Continuum Clock Definition

The Continuum Clock $T(x)$: We define a continuous scalar clock field $T(x)$ whose gradient is the coarse-grained event current: $\partial_\mu T(x) \equiv (1/\nu_0)(j_{\mu\text{ev}})^\ell$

$$\text{IR Equivalence: } N_{\text{ev}}(\Sigma) \approx \nu_0 \int_\Sigma d\Sigma_\mu \partial_\mu T = \nu_0 \Delta T$$

Constructing the Relational TOA Distribution

The probability density $p(\Theta)$ for a particle to arrive at a clock time Θ is defined by binning the matter flux $F(\sigma)$ according to the value of the clock field $T(x)$ on the detector surface.

$$p(\Theta) \equiv \frac{1}{N} \int_{\Sigma} d^3\sigma \sqrt{|h|} F(\sigma) \delta(T_{\Sigma}(\sigma) - \Theta)$$

“Integrate the incoming matter flux over the entire detector surface...”

“...but only count contributions where the physical clock on the detector, $T_{\Sigma}(\sigma)$, reads the specific time Θ .”

“Normalize the result to get a probability.”

The Framework is Covariant, Unobstructed, and Physically Grounded



Manifestly Covariant: The definition of $p(\Theta)$ is built from scalars and tensors, respecting the principles of relativity from the ground up.

$$\cancel{[T, H] = i}$$

Free of Pauli-type Obstructions: By defining TOA via field data on a hypersurface, we never need to introduce a problematic time operator conjugate to the Hamiltonian.



Detector-Independence: The formalism is independent of detector microphysics. The only inputs are the detector's geometry (Σ) and the dynamics of the clock sector, which are themselves part of the physical theory.

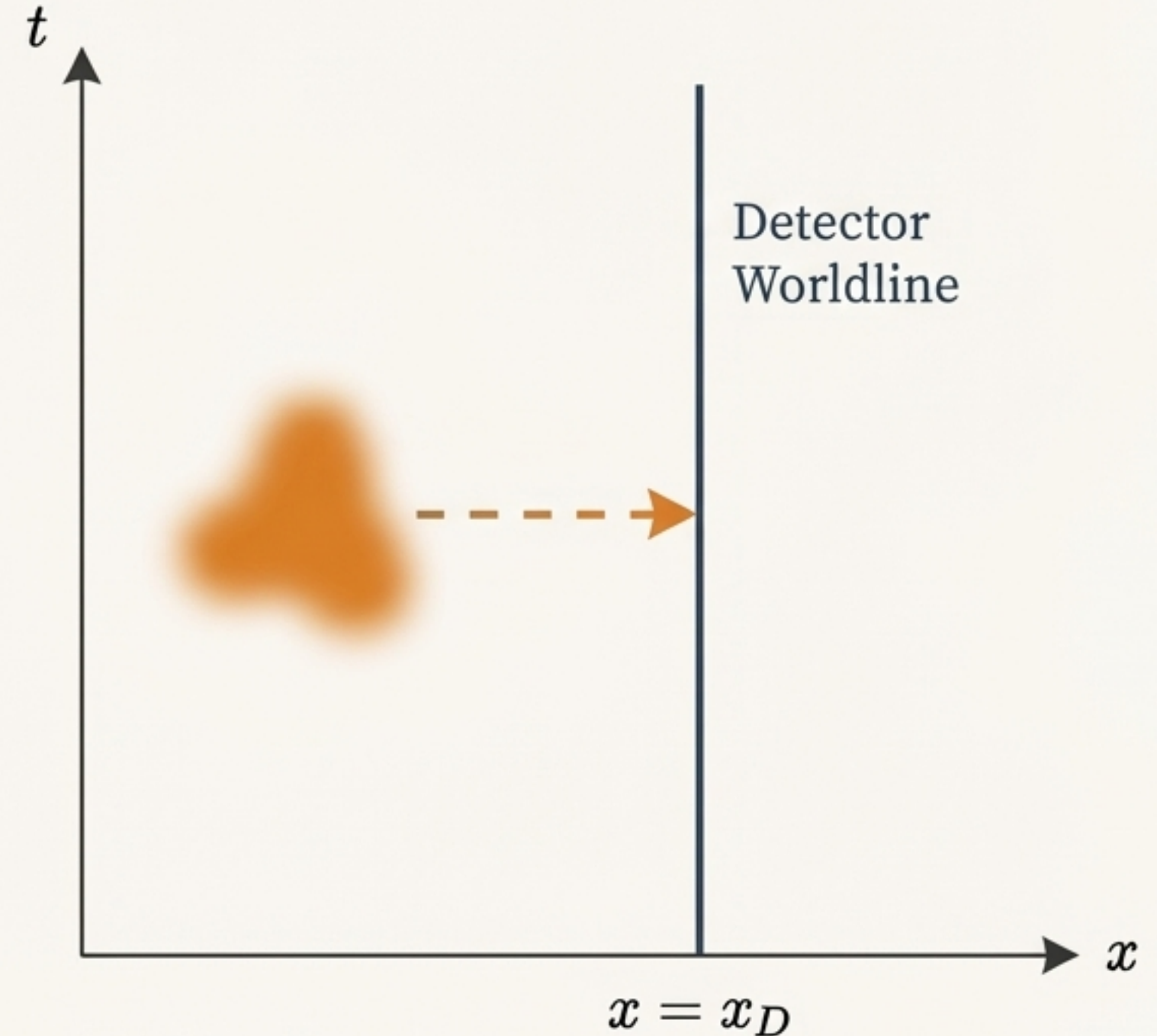
A Worked Example: Does the Framework Yield Sensible Predictions?

The Setup (1+1 Dimensions)

- Spacetime: 1+1D Minkowski space (t, x) .
- Detector: A stationary worldline at a fixed position $x = x_D$.
- Particle: A complex Klein-Gordon field ψ of mass m .
- Matter Current: The standard U(1) conserved current, $J^\mu = \frac{i}{2m}(\psi^* \partial_\mu \psi - \psi \partial_\mu \psi^*)$.

Goal

Compute the TOA distribution $p(\Theta)$ for an incoming wavepacket and a fluctuating clock, and check if the result is physically reasonable.

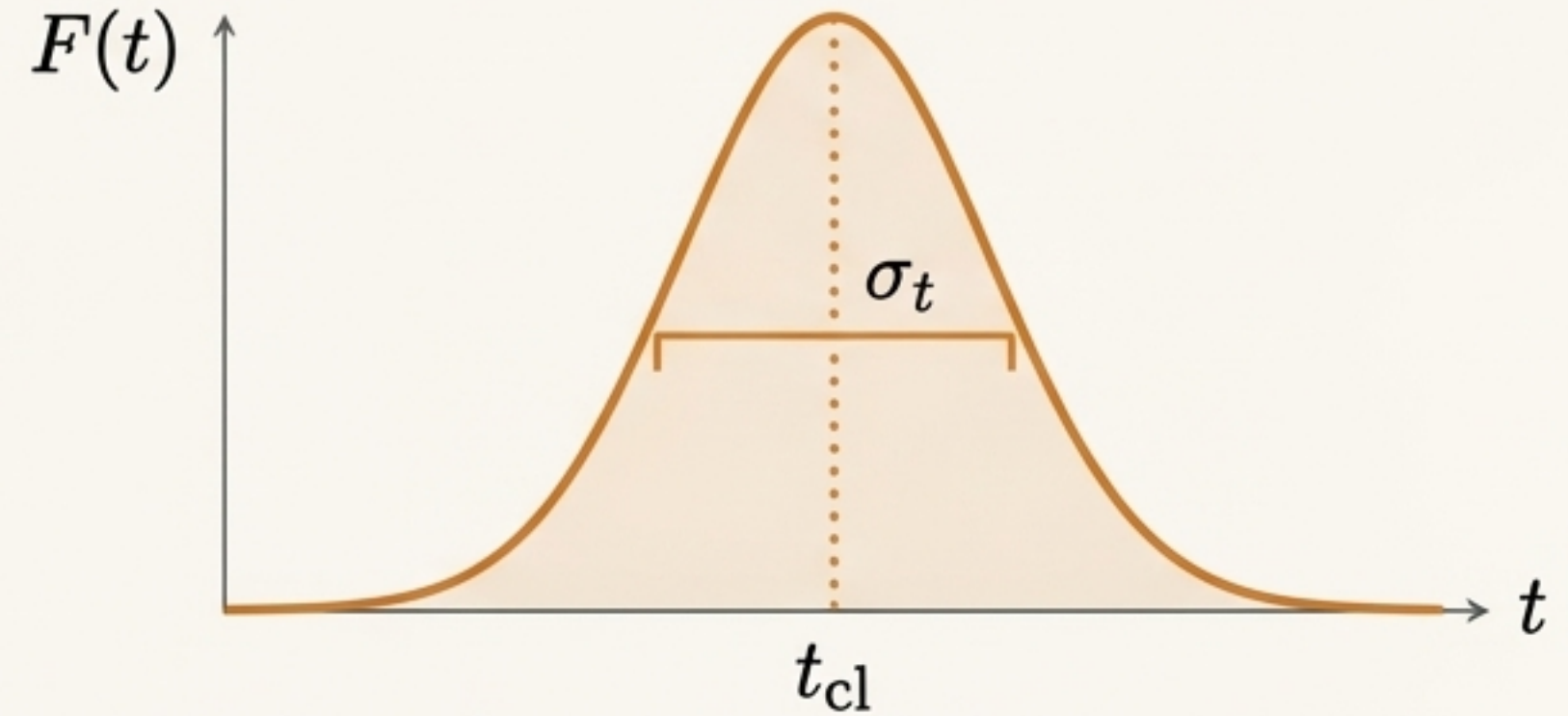


Specifying the Inputs: A Gaussian Wavepacket and a Fluctuating Clock

Input 1: The Semiclassical Matter Flux

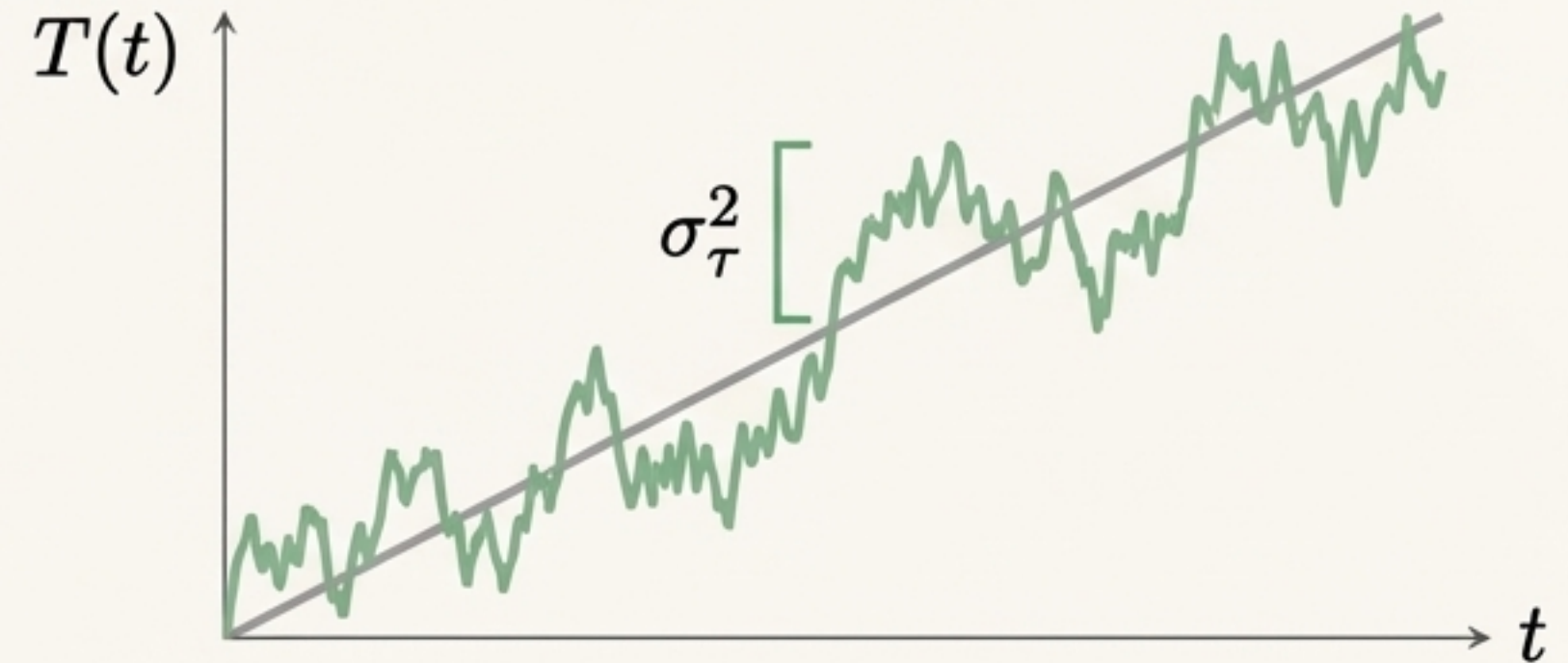
We start with a wavepacket $\psi(t, x)$ that is narrowly peaked in momentum. This results in an approximately Gaussian arrival profile for the flux at the detector:

$$F(t) \approx F_0 \exp \left[-(t - t_{cl})^2 / 2\sigma_t^2 \right]$$



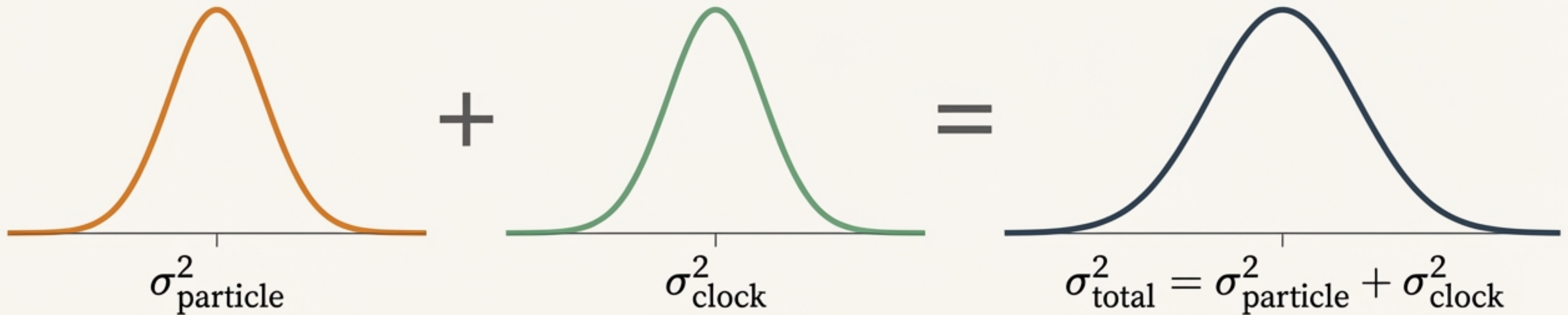
Input 2: The Fluctuating Clock Field

The clock field on the detector is modeled as $T(t) = t + \tau(t)$, where $\tau(t)$ represents physical fluctuations. We assume $\tau(t)$ is a stationary Gaussian random variable with mean zero and variance $\sigma_\tau^2 \equiv \langle \tau(t)^2 \rangle$



The Result: The Final Distribution is a Convolution of the Inputs

$$p(\Theta) = \frac{1}{\sqrt{2\pi(\sigma_t^2 + \sigma_\tau^2)}} \exp \left[-\frac{(\Theta - t_{cl})^2}{2(\sigma_t^2 + \sigma_\tau^2)} \right]$$

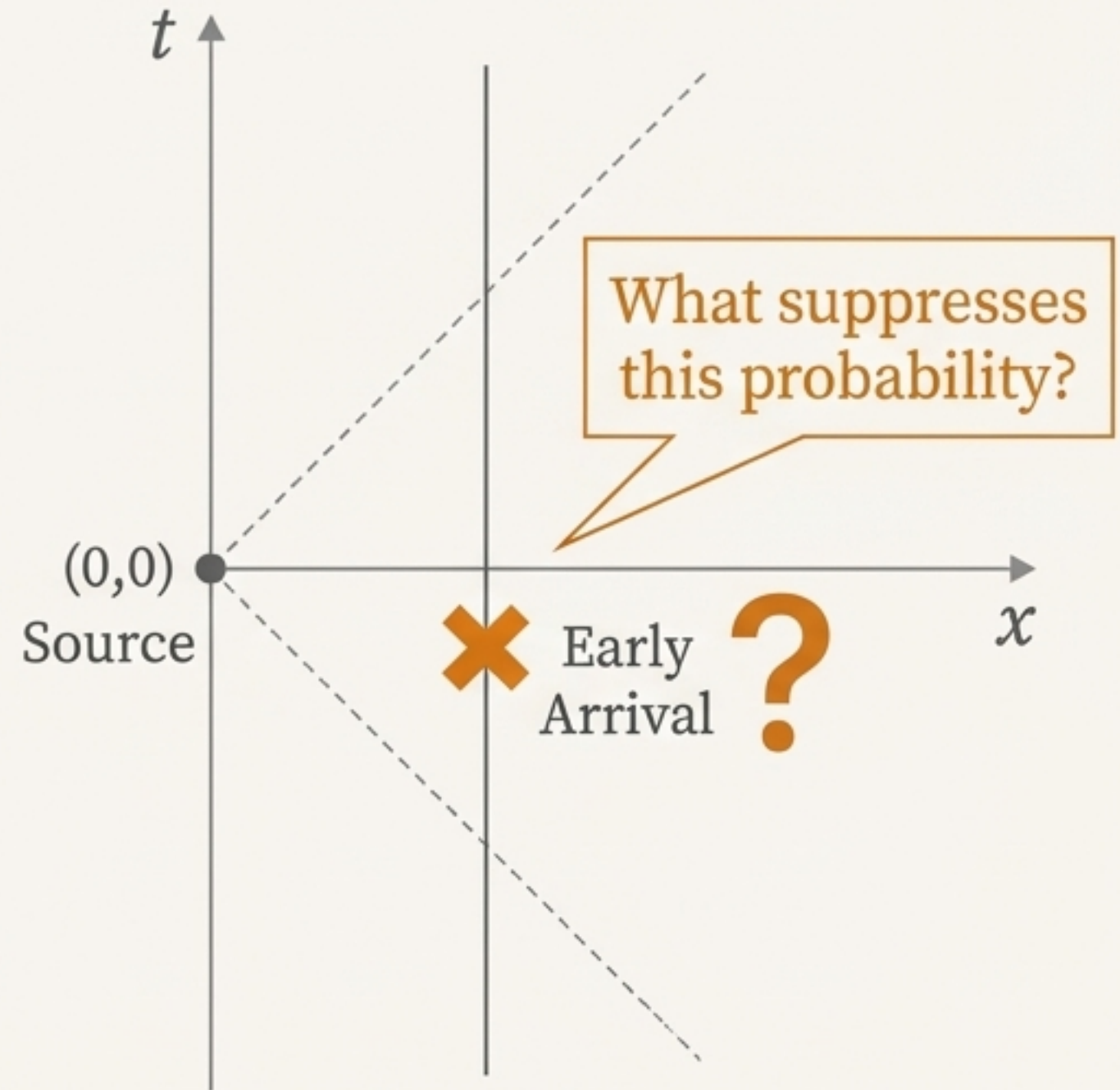


Predictive Broadening: The total variance of the measured arrival time is the sum of the variances of the inputs. $\text{Var}(\Theta) = \sigma_{\text{total}}^2 = \sigma_{\text{particle}}^2 + \sigma_{\text{clock}}^2$

Beyond Broadening: The Clock Also Suppresses Acausal Arrivals

A Deeper Question

- What prevents the detector from registering an arrival “impossibly” early? For example, faster than the speed of light would permit?
- In our framework, such events are not forbidden outright but are dynamically suppressed.
- The suppression is not due to the matter field ψ alone, but arises from the correlation properties of the physical clock field $\tau(x)$.
- An ‘early’ arrival would require a strong correlation in the clock field between the particle’s source and the detector across a spacelike interval. The clock’s own dynamics prevent this.



Mechanism: A Massive Clock Field Leads to Exponential Suppression

Clock Field Dynamics

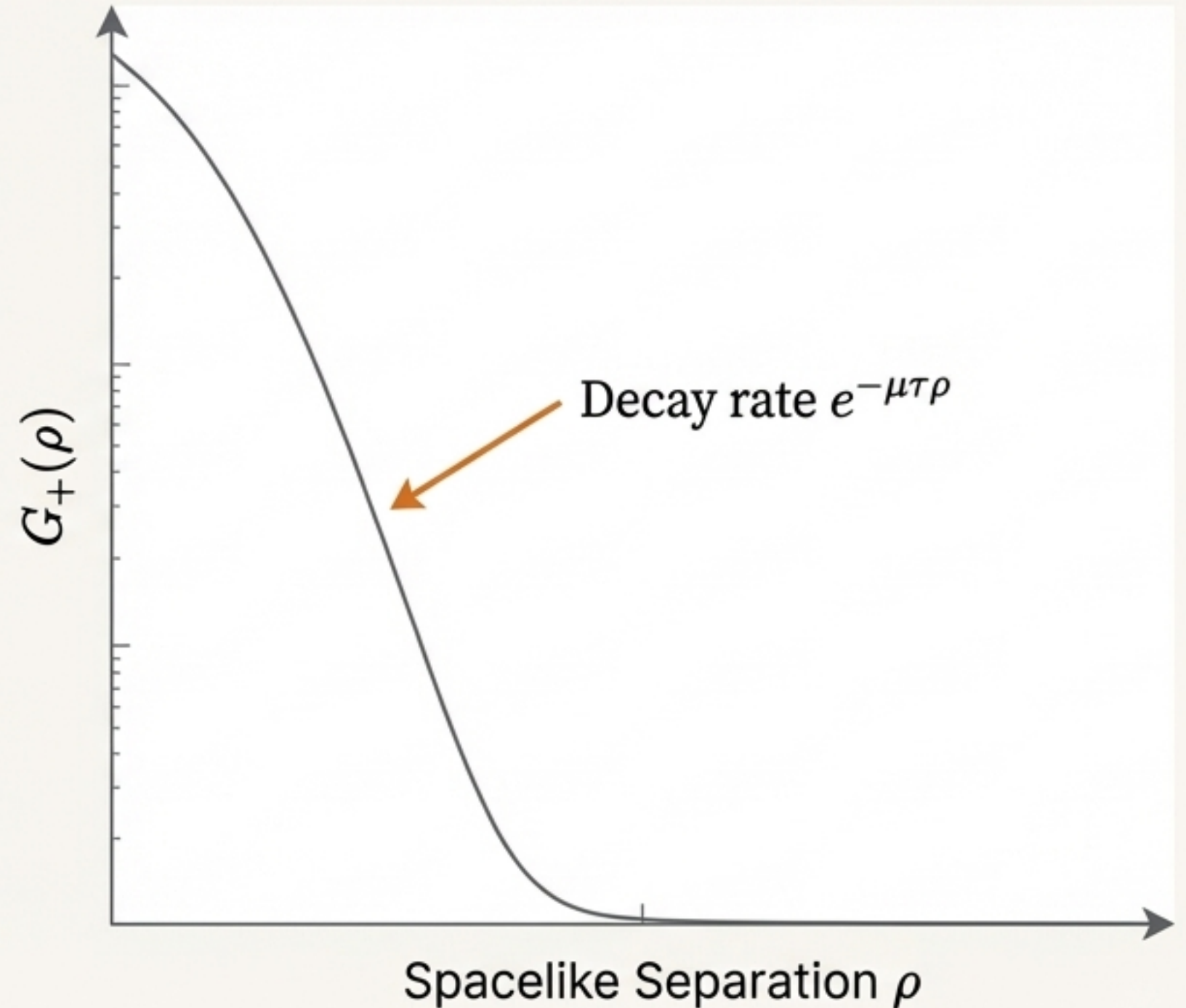
- The clock fluctuations $\tau(x)$ are governed by an effective field theory (EFT). The simplest stable model is that of a massive scalar field with mass μ_τ .
- The two-point correlation function $G_+(\Delta t, \Delta x)$ for a massive field decays exponentially for spacelike separations.

The Asymptotic Result:

$$G_+(\rho) \sim \frac{1}{\sqrt{\rho}} e^{-\mu_\tau \rho}, \quad \rho = \sqrt{(\Delta x)^2 - (\Delta t)^2}$$

- **The Implication:**

Any process depending on such a correlation is **exponentially suppressed**. The clock's mass μ_τ sets a physical scale that enforces causality.



An Example Realization: A Clock from Counting Defect Worldlines

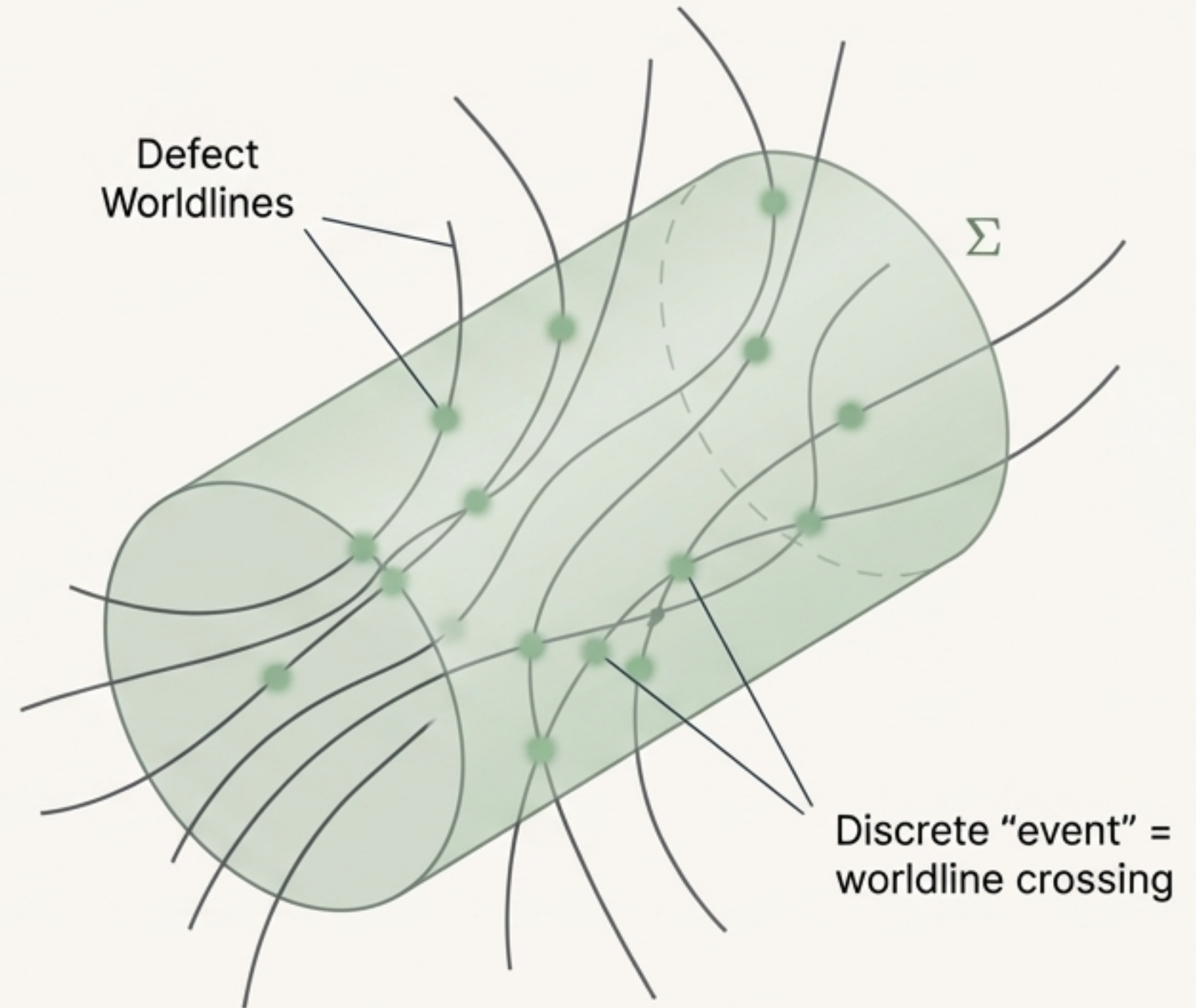
A Concrete Model

- Where could an “event current” come from? Consider a system with stable, localized defects (e.g., topological defects, vortices).

- The Event Current $j^{\mu ev}$:

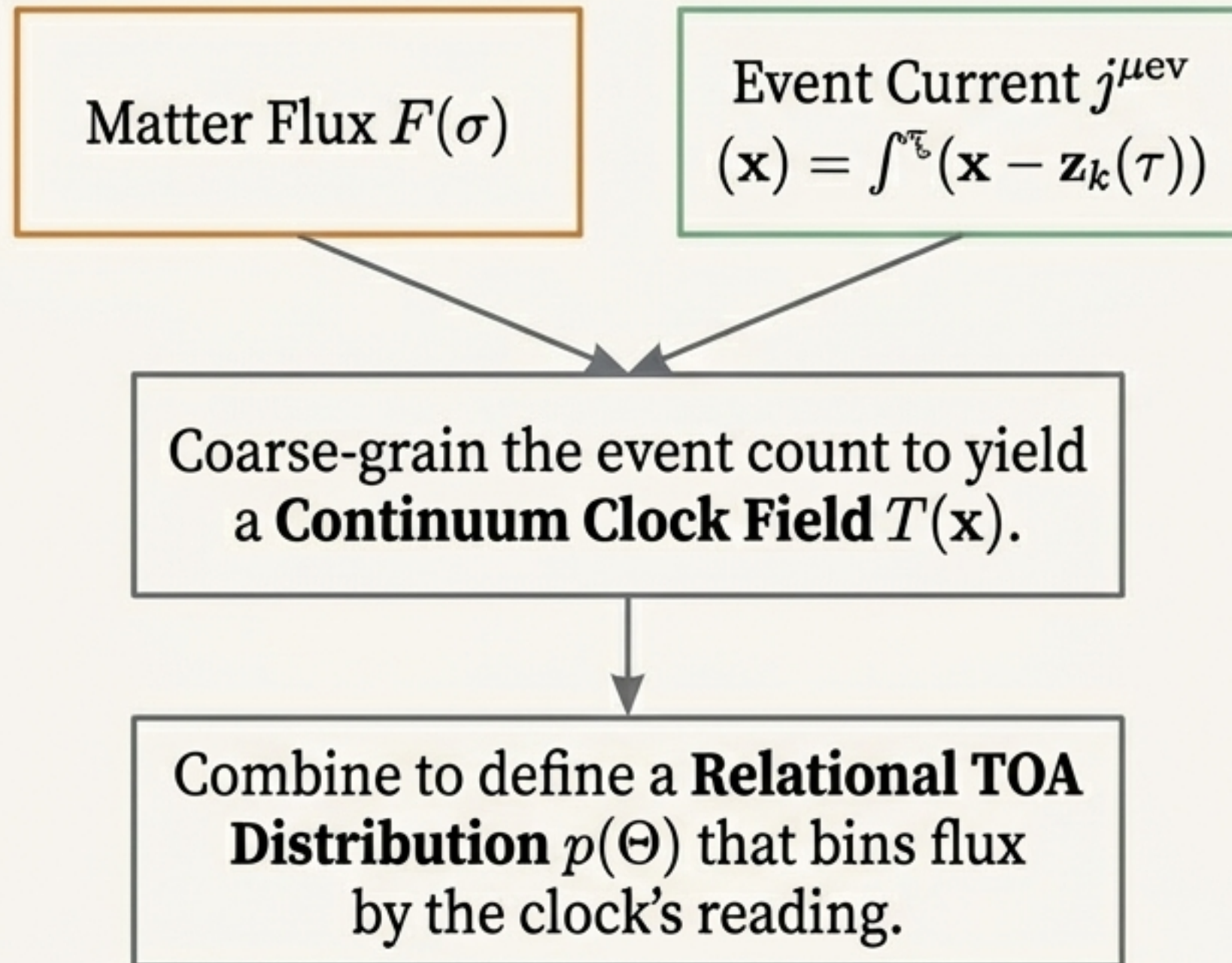
$$j^{\mu ev}(\mathbf{x}) \equiv \sum_k q_k \int d\tau \left(\frac{dz_k^\mu}{d\tau} \right) \delta^4(\mathbf{x} - \mathbf{z}_k(\tau))$$

- The Event Count N_{ev} : In this model, N_{ev} simply counts the net number of defect worldlines that pierce the detector world-tube.
- The Continuum Clock $T(\mathbf{x})$: The coarse-grained version of this defect current gives a smooth clock field, which can be interpreted as a physical foliation of spacetime.



Summary: A Relational TOA Built from Physical Currents

Recap of the Construction

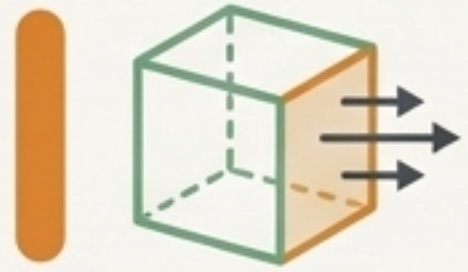


Key Results

- The framework is **covariant** and avoids Pauli's obstruction.
- It produces **predictive broadening**: $\text{Var}(\Theta) = \sigma_{\text{particle}}^2 + \sigma_{\text{clock}}^2$.
- It provides a physical mechanism for the **exponential suppression** of acausal arrivals, governed by the clock mass $\mu\tau$.

The Path Forward: A Falsifiable Program for TOA Statistics

This construction is not just a definition, but a program for making quantitative, falsifiable predictions about TOA statistics across different resolutions.



1. Extend to 3+1D: Perform a worked example for a finite-size detector to quantify how TOA statistics depend on detector geometry.



2. Beyond Gaussian Clocks: Investigate non-Gaussian clock dynamics (beyond the simple quadratic EFT) to predict higher-order signatures (skew, kurtosis) in the $p(\Theta)$ distribution.



3. Model UV Corrections: Systematically describe the ‘renewal-type’ corrections to $p(\Theta)$ that appear at very high time resolution ($\Delta t \lesssim \ell$), modeling the crossover to the continuum limit.